

Woojun Jung

 [woojoonjung.github.io](https://github.com/woojoonjung) |  woojoon@korea.ac.kr |  Google Scholar |  LinkedIn |  GitHub

RESEARCH INTERESTS

Representation Learning: learning representations from heterogeneous data and interaction histories for adaptive memory and long-horizon personalization.

Human-AI Interaction: designing conversational AI that helps people articulate their needs, understand information, and make informed decisions.

EDUCATION

Korea University

M.S. in Computer Science and Engineering

Advisor: Prof. [Susik Yoon](#)

Mar 2025 – Feb 2027 (Expected)

Korea University

B.E. in Computer Science and Engineering

Mar 2022 – Feb 2025

Korea University

B.F.A. in Industrial and Information Design

2 years of Military Service: Republic of Korea Air Force (2017 – 2019)

Mar 2016 – Feb 2022

Relevant Coursework: *CS/AI* - Algorithms (A+), Computer Architecture (A), Artificial Intelligence (A), Deep Learning (A), Optimization Theory (A), Reinforcement Learning (A), Natural Language Processing (A), Brain and Machine Cognition (A+); *Design/HCI* - Interaction Design (A+), Interface Design (A+), Design and Human Factors (A+), Design Methodology (A+).

PUBLICATIONS

**: equal contribution*

- [3] **Segment-driven Structural Induction and Semantic Alignment for Heterogeneous Tabular Representation**
[Woojun Jung](#) and [Susik Yoon](#)
ICML 2026 — [\[arXiv\]](#) [\[Code\]](#)
- [2] **Can Structural Cues Save LLMs? Evaluating Language Models in Massive Document Streams**
[Yukyung Lee](#), [Yebin Lim*](#), [Woojun Jung*](#), [Wonjun Choi](#), [Susik Yoon](#)
KDD 2026 — [\[ACM DL\]](#) [\[Code\]](#)
- [1] **HAETAE: In-Domain Table Pretraining with Header Anchoring**
[Woojun Jung](#) and [Susik Yoon](#)
SIGIR 2025 (Short) — [\[ACM DL\]](#) [\[Code\]](#)

RESEARCH EXPERIENCE

DAIS Lab, Korea University

Research Intern (Aug 2024 – Feb 2025); Research Assistant (Mar 2025 – Present)

Aug 2024 – Present

Calibrated User Beliefs for Conversational Agents

May 2026 – Present

- Formulating long-horizon personalization as maintaining calibrated beliefs over context-dependent and evolving user preferences rather than reducing them to fixed textual point memories.
- Designing pilot comparisons across deterministic text memories and probabilistic preference states and preference states and probabilistic preference states.
- Exploring whether representation-derived uncertainty can be calibrated against subsequent user feedback and used to guide when agents should apply a preference or ask for clarification.

Structured Understanding in Evolving Information Streams

KDD 2026

- Developed the Temporal QA task, one of three benchmark tasks alongside Topic Clustering and Summarization.
- Designed and implemented its evaluation protocol, benchmark dataset, and human-annotation guidelines.

Structured Understanding of Tabular Data

ICML 2026; SIGIR 2025

- Led two first-author projects on domain-specific tabular representation learning.
- Proposed header anchoring to regularize contextualized header embeddings toward context-independent anchors, stabilizing semantics across row contexts and schema variants.
- Developed a pretraining method that jointly learns schema and column contexts with entropy-routed objectives to preserve domain-coherent and instance-specific attribute semantics.

APPLIED RESEARCH COLLABORATION

Foundation Models for Enterprise Relational Data, KT (Korea Telecom)

Mar 2025 – Present

- Collaborated with KT on foundation-model methods for enterprise relational databases.
- Analyzed challenges in adapting tabular ML to organizational data with heterogeneous schemas.
- Prototyped a pretraining pipeline for schema-aware attribute representations for enterprise tables.
- Investigating schema-linking methods that map natural-language questions to relevant tables and columns.

SELECTED PROJECTS

Dr.Snap: Multimodal Dermatology Decision-Support System

2024

- Designed and built a conversational prototype that combines skin photos and symptom descriptions to provide preliminary guidance before a clinic visit.
- Developed a dialogue workflow that uses retrieved condition information and prior user responses to generate follow-up questions and narrow candidate conditions.

Bianca: Full-Scale Interactive Refrigerator Prototype

2020

- Developed Arduino code and electronics and built a full-scale refrigerator prototype that maps handle-press locations to corresponding illuminated regions on the door.

TEACHING EXPERIENCE

Teaching Assistant, Data Science, Korea University

Spring 2025

- Prepared and delivered tutorials on LaTeX and visualization with matplotlib for undergraduate students.

Teaching Assistant, Artificial Intelligence, Korea University

Fall 2025

- Prepared and delivered a tutorial on multimodal representation learning with CLIP for undergraduate students.

AWARDS & SCHOLARSHIPS

Master's Fellowship (KRW 12,000,000), National Research Foundation of Korea (NRF)

2025–2026

Third Place, Korea University Capstone Design Competition

2024

Third Place, Samsung Design Competition

2019

SKILLS

Programming: Python, C, SQL, Git, GitHub, LaTeX, HTML, CSS, JavaScript

ML & Systems: PyTorch, CUDA, Transformers, vLLM, NumPy, pandas, scikit-learn, matplotlib

Design & Hardware Prototyping: Figma, Illustrator, Photoshop, Rhinoceros, KeyShot, 3D printing, Arduino

Languages: Korean (Native), English (Fluent)